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Dipole Moment of Lithium Hydride

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(Received August 8, 1960)

WE have measured the electric dipole moment of lithium hydride by the molecular beam electric resonance method.[1] Our apparatus utilizes electric quadrupole A and B fields for state selection and space focusing of the beam.[2] The uniform electric C field transition region is 15 cm long. The transition observed is the rotational angular-momentum reorientation $J=1, m_J=0 \rightarrow J=1, |m_J|=1$. For Li^7H in the ground-vibrational-state this transition was observed at about 3.2 Mc in an electric field of 730 v/cm, 22.7 Mc at 1920 v/cm, and 63.6 Mc at 3280 v/cm. The lines arising from the hyperfine structure of this transition extend over a range of 240 kc and the minimum width of these lines is 7 kc at optimum rf power. The hyperfine line pattern is identical at the three observed frequencies, with each line showing the expected strong field behavior

$$\nu_i(E)=\nu_{\text{STARK}}(E)+\alpha_i,$$

where $\nu_{\text{STARK}}(E)$ is the unperturbed Stark frequency and α_i is the voltage-independent displacement of a line from this frequency.

The electric dipole moments of Li^7H and Li^6H are[3]

$$\begin{aligned} Li^7H \quad V=0 \quad & \mu=5.882\pm0.003D \\ V=1 \quad & \mu=5.990\pm0.003D \\ Li^6H \quad V=0 \quad & \mu=5.884\pm0.003D. \end{aligned}$$

This research has been aided by support from the Atomic Energy Commission and the Research Corporation.

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[1] N. F. Ramsey, *Molecular Beams* (Oxford University Press, New York, 1956); P. Kusch and V. Hughes, *Encyclopedia of Physics*, (Springer-Verlag, Berlin, Vienna, 1959) Vol. 37.

[2] G. Bennewitz, W. Paul, and C. Schlier, Z. Physik **141**, 6 (1955); J. P. Gordon, H. J. Zeiger, and C. H. Townes, Phys. Rev. **99**, 1264 (1955).

[3] The rotational constants of Li^7H, $V=0$ and $V=1$, are the effective values of F. H. Crawford and T. Jorgensen, Phys. Rev. **47**, 932 (1935), while that of Li^6H is calculated from these constants.